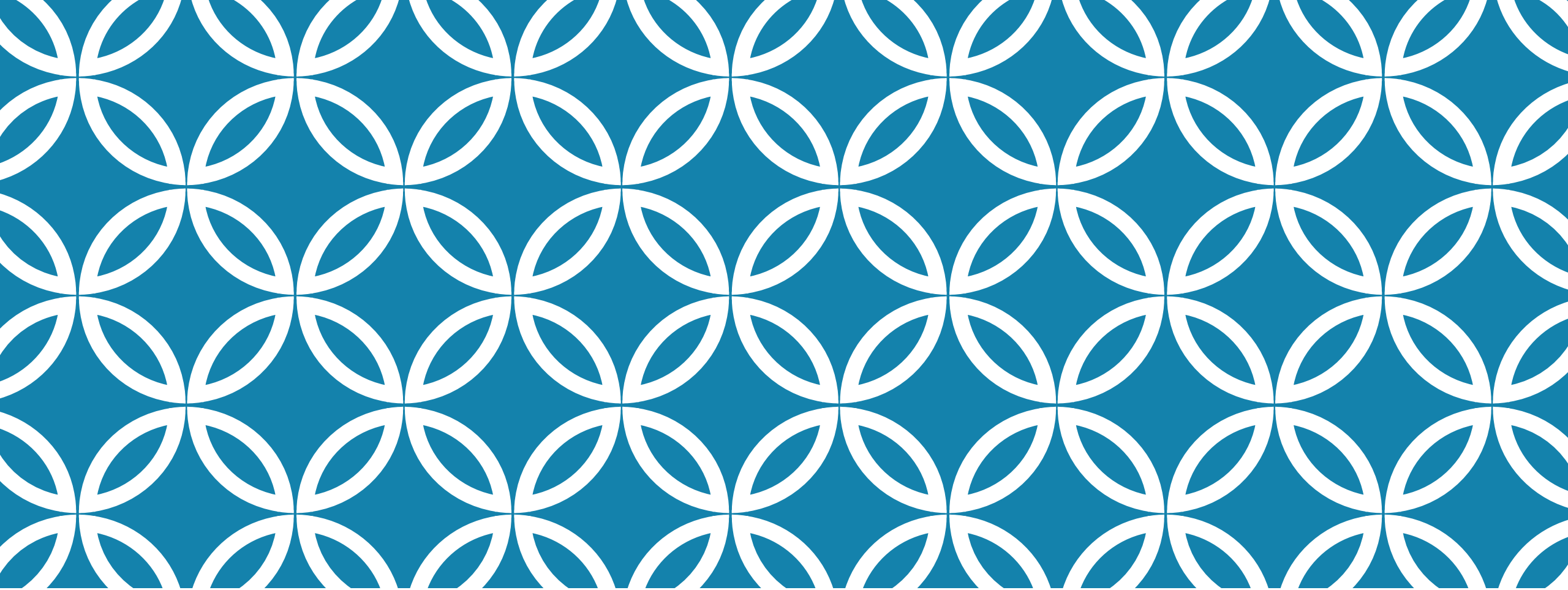


PRACTICE QUESTIONS

Probability Distribution

34. A Harris Interactive survey for InterContinental Hotels & Resorts asked respondents, “When traveling internationally, do you generally venture out on your own to experience culture, or stick with your tour group and itineraries?” The survey found that 23% of the respondents stick with their tour group (*USA Today*, January 21, 2004).

a. In a sample of six international travelers, what is the probability that two will stick with their tour group?

b. In a sample of six international travelers, what is the probability that at least two will stick with their tour group?

c. In a sample of 10 international travelers, what is the probability that none will stick with the tour group?

In San Francisco, 30% of workers take public transportation daily (*USA Today*, December 21, 2005).

a. In a sample of 10 workers, what is the probability that exactly 3 workers take public transportation daily?

b. In a sample of 10 workers, what is the probability that at least 3 workers take public transportation daily?

When a new machine is functioning properly, only 3% of the items produced are defective. Assume that we will randomly select two parts produced on the machine and that we are interested in the number of defective parts found.

a. Describe the conditions under which this situation would be a binomial experiment.

b. Draw a tree diagram similar to show this problem as a two-trial experiment.

c. How many experimental outcomes result in exactly one defect being found?

d. Compute the probabilities associated with finding no defects, exactly one defect, and two defects.

A Randstad/Harris interactive survey reported that 25% of employees said their company is loyal to them (*USA Today*, November 11, 2009). Suppose 10 employees are selected randomly and will be interviewed about company loyalty.

a. Is the selection of 10 employees a binomial experiment? Explain.

b. What is the probability that none of the 10 employees will say their company is loyal to them?

c. What is the probability that 4 of the 10 employees will say their company is loyal to them?

d. What is the probability that at least 2 of the 10 employees will say their company is loyal to them?

Military radar and missile detection systems are designed to warn a country of an enemy attack. A reliability question is whether a detection system will be able to identify an attack and issue a warning. Assume that a particular detection system has a .90 probability of detecting a missile attack. Use the binomial probability distribution to answer the following questions.

a. What is the probability that a single detection system will detect an attack?

b. If two detection systems are installed in the same area and operate independently, what is the probability that at least one of the systems will detect the attack?

c. If three systems are installed, what is the probability that at least one of the systems will detect the attack?

d. Would you recommend that multiple detection systems be used? Explain.

Twelve of the top 20 finishers in the 2009 PGA Championship at Hazeltine National Golf Club in Chaska, Minnesota, used a Titleist brand golf ball (GolfBallTest website, November 12, 2009). Suppose these results are representative of the probability that a randomly selected PGA Tour player uses a Titleist brand golf ball. For a sample of 15 PGA Tour players, make the following calculations.

a. Compute the probability that exactly 10 of the 15 PGATour players use a Titleist brand golf ball.

b. Compute the probability that more than 10 of the 15 PGA Tour players use a Titleist brand golf ball.

c. For a sample of 15 PGA Tour players, compute the expected number of players who use a Titleist brand golf ball.

A university found that 20% of its students withdraw without completing the introductory statistics course. Assume that 20 students registered for the course.

a. Compute the probability that 2 or fewer will withdraw.

b. Compute the probability that exactly 4 will withdraw.

c. Compute the probability that more than 3 will withdraw.

d. Compute the expected number of withdrawals.

According to a survey conducted by TD Ameritrade, one out of four investors have exchange-traded funds in their portfolios (*USA Today*, January 11, 2007). Consider a sample of 20 investors.

a. Compute the probability that exactly 4 investors have exchange-traded funds in their portfolios.

b. Compute the probability that at least 2 of the investors have exchange-traded funds in their portfolios.

c. If you found that exactly 12 of the investors have exchange-traded funds in their portfolios, would you doubt the accuracy of the survey results?

d. Compute the expected number of investors who have exchange-traded funds in their portfolios.